

Exploring the usability of SeeSaw, an educational kiosk-based mobile application

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Abstract. SEESAW is a mobile kiosk-based app that aims to teach research ethics and integrity principles to students and researchers. The app incorporates various interactive components such as polls, and sorting activities, which aim to engage its users via two decision-making scenarios. This paper evaluates the usability of the app using a mixed-methods approach. This combines USE (Usefulness, Satisfaction, Ease of Use) questionnaires along with a heuristic evaluation and qualitative feedback from users. Our findings reveal that participants perceived SEESAW positively with a high overall rating in terms of learnability, ease-of-use, and user satisfaction. User feedback has also highlighted a few issues with components that violated several heuristics and which could be improved – such as providing more use control, better consistency, and the introduction of better error recovery mechanisms. These findings underscore the need for balancing the guided nature of learning activities with providing more control and freedom to users. Finally, the study offers several insights for the design of kiosk-based applications which can help developers design better interactive experiences within educational contexts.

Keywords: Usability · Kiosk · App · Heuristics · User experience

1 Introduction

The use of *Self-Service Kiosks* (SSKs) in commercial and organizational environments has become a part of everyday life [9,17]. The ubiquitous nature of these devices makes them perfect for applications such as advertising, information points, ordering systems, and also as platforms serving educational purposes [12,17]. SSKs allow people to access information and receive assistance without other human intervention. In return, this reduces labor costs and effectively streamlines operations for various use cases. At the same time, the usability of the kiosks is a critical aspect determining their effectiveness in delivering information or enabling the use of commercial services [12]. Therefore, poor design decisions can negatively impact the *User eXperience* (UX), leading to confusion, frustration, and in response to reduced adoption.

This study builds on work which explores the usability of both kiosk and mobile apps, and which reviews several aspects of usability evaluation. This includes

heuristic evaluation methods, their effectiveness, and the aspect of conducting a usability evaluation on mobile applications [16,17]. An additional aspect in terms of evaluating kiosk and other types of mobile apps is *context-sensitive* evaluation [7,20]. Given that kiosk apps are frequently utilized in a variety of environments and circumstances – such as crowded or noisy rooms – it is important to carefully consider the physical and environmental factors that affect their performance [5].

This paper presents and evaluates SEESAW, a kiosk-based educational mobile application developed using the Flutter framework. The SEESAW app aims to educate learners about research ethics and integrity by presenting interactive scenarios with various types of multimedia and interactive features such as videos, polls, sorting activities, etc. A mixed methods approach is used to evaluate the usability of SEESAW, combining both qualitative and quantitative data. The study primarily employs the standardized *Usefulness, Satisfaction, Ease of use* (USE) questionnaire [1,11] to uncover the advantages and disadvantages of the app’s user interface design and to answer the following research questions:

- RQ1:** What are the main usability challenges in designing educational kiosk-based applications?
- RQ2:** How effective is SEESAW’s User Interface (UI) in enabling a good level of task performance?
- RQ3:** How do users experience the SEESAW app and how do they perceive its usability?

The study uses questions based on Nielsen’s heuristics [4,14] to evaluate the level of usability with regards to different aspects of the app. A set of participants was involved in our experiment to respond to these questions, reporting any issues they faced with the app, as well as providing any written suggestions. The responses of the participants were recorded and then analysed using descriptive statistics to uncover the strengths and limitations of SEESAW and to answer the aforementioned research questions.

The rest of this paper is organised as follows. Section 2 discusses the background and literature relevant to the development and evaluation of kiosk-based applications. Section 3 describes the design decisions and implementation of the SEESAW application. Section 4 outlines our methodology and experimental design which we used to evaluate its usability, and section 5 presents the data we collected in the usability experiments. Next, in section 6 we analyse the data through a mixed-methods analysis, and we juxtapose our findings to the original research questions. Finally, section 7 concludes the paper with a discussion of the impact of these findings, and offers recommendations for improving the usability of kiosk-based applications.

2 Background & Related Work

SSKs are an integral part of consumer-focused economies, serving as potential hubs for product advertising, service delivery such as self-service ordering, and

informational purposes such as healthcare, transportation. They are used in education, albeit less frequently. These systems aim to utilise interactive features to provide a high level and seamless experience for their users. Therefore, their success largely depends on the level of user experience (UX) they can achieve. Overall, it is accepted that when the design and engineering of such systems does not achieve an adequate UX, there is a negative impact which is translated to weaker user adoption [6,15].

2.1 Usability Evaluation Methods

The usability evaluation of SSKs is a multifaceted area involving both quantitative and qualitative measurements. One of the most common methods used is a *heuristic evaluation*, which evaluates an app with respect to a set of heuristics such as Nielsen’s [14]. Other methods are also used, such as *user observations*, *task-based assessment*, *expert analysis*, *thinking-aloud protocols*, and more. In the past, these have been used to measure the usability of SSKs in order to uncover potential issues in their design [3,8]. Furthermore, standardized approaches also exist to evaluate usability in a variety of domains, which can include SSKs. For example, the USE questionnaire provides a standardized and structured way to assess the users’ perceptions of a product [10,18] with respect to the stated factors abbreviated in its name. Such standardised approaches can provide valuable insights both during the iterative design processes, but is also used to evaluate fully developed prototypes [18].

2.2 Usability in Specialized Domains

Although the field of usability is very broad, our work focuses on the usability of kiosk-based applications, which has significant differences from other domains. Therefore, a specialised evaluation process is required which is adjusted to the unique requirements of this specialised domain. As an example, usability studies done in the fast-food industry domain revealed that order accuracy and transaction times are directly related to the design of the interface of the kiosk [13]. *User Interface* (UI) elements and layouts (e.g. menu layouts) and a clear design hierarchy and call-to-action components can positively contribute to a better experience by reducing the number of steps and thus reduce cognitive load, which in result improves sales. Other studies have also revealed that a good interface structure can reduce user frustration, especially among first-time adopters or those with minimal experience in touchscreen interactions [2]. This is also relevant in the healthcare, transportation, and education domains. However, healthcare institutions operate under very different protocols from those used in restaurants. For instance, hospitals use such apps for patient check-ins, appointment scheduling, and so on, which must account for a very diverse user demographic – such as the elderly, those with limited experience using technology, etc. With respect to this, a study has shown that there is a need for accessibility features to be implemented in SSKs [19]. Features such as large buttons, text-to-speech technology, multilingual support, and high-contrast text

can improve the accessibility of the system and provide a higher level of usability for all its users. Altogether, these examples clearly indicate the need to conduct a *domain-specific* usability evaluation to ensure that, with respect to our domain, the SSKs employed in an educational setting are primed to provide the best level of user experience in that particular setting.

3 App Design & Implementation

The SEESAW app was developed as a cross-platform, kiosk-based application using the Flutter framework and the Dart programming language. Initially, a set of wireframes was created as an early prototype of the app idea. This enabled brainstorming and idea generation. The SEESAW app includes three types of different activities, two of which are interactive. Initially, users must press the Start button and select one of two different perspectives: *Policy Maker* or *Research Ethics Committee Member*. Users experience different content based on the type of perspective they have chosen, which can impact their decision-making process.

A *video page*, as shown in Figure 1, guides users to watch videos relevant to the perspective they have chosen. Depending on the selected perspective, users will watch different videos throughout their experience which may influence how they make ethical decisions. Users which are already familiar with the content of the video can optionally skip to the next section.

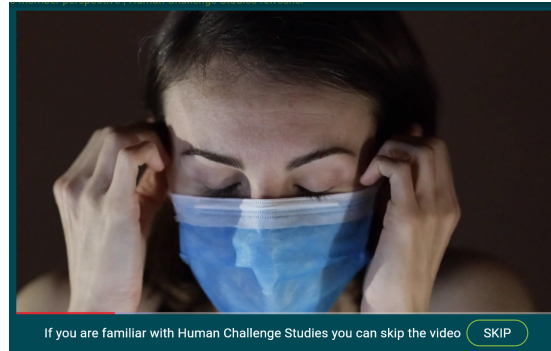


Fig. 1: The video page user interface.

In addition to videos, users are also asked to complete short *decision-making sorting activities* where they are shown items resembling movable, coloured rectangles. These contain keywords or statements and the users must decide where they should be categorised based on their opinion. Each sortable statement is shown as a selectable and draggable component in the middle of the screen, as shown in Figure 2. On either side of the screen, there are two boxes that correspond to two categories or classifications. In the example of Figure 2 these are

called “Pros” and “Cons”, guiding users to place the items which they think are *positive* or *advantageous* in the “Pros” box, and the ones they think are *negative* or *non-advantageous* in the “Cons” box. This is done by long-pressing on each item as instructed until it is picked up, and then dragging it to the corresponding box.

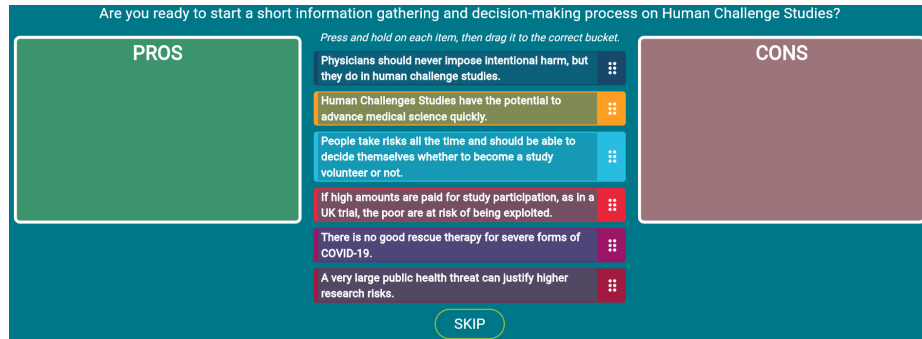


Fig. 2: The sorting activity user interface.

The third type of activity is a *poll/voting* activity where users are asked to provide their opinion about an ethical decision using a “Yes” or “No” answer. The learners are presented with a question and they must make a decision by first considering the statement and then responding by pressing the “Yes” or “No” button, as shown in Figure 3. Once users respond, the app shows the initial results of the poll, as shown in Figure 4a. These include the answers of all users and indicate how many users have responded with either of the two options. In the next step of this process, users are then shown an informative video regarding the ethical question and are again asked to provide their answer to the same question using the same component. This is of special interest, as the additional information could possibly cause the learners to change their mind. The app then shows a comparison of the two decisions, quantifying changes between the two options from the first to the second vote. For instance, if 10 learners initially voted “No”, but 5 of them changed their answer to “No” in their second vote, the app will show that 5 users changed their answer from “Yes” to “No”, and the opposite, as demonstrated in Figure 4b.

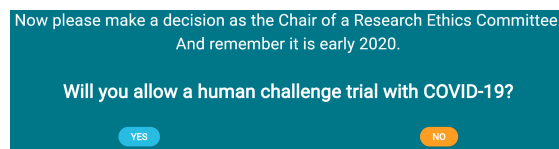


Fig. 3: The poll decision user interface.



(a) The poll results user interface before making a second consideration (b) The poll results after making a second consideration.

Fig. 4: Comparison of poll results before and after a second consideration.

3.1 Kiosk-specific features

The SEESAW app is designed for kiosk devices with high-definition displays in landscape mode. Several devices were assessed prior to deployment, considering factors such as screen size, orientation, weight, and price. The selected devices run Android 11 and include WiFi adapters to access required online content, such as videos and poll services.

Design decisions for SEESAW assume ample screen space in landscape orientation for displaying components and supporting touch interaction. Several kiosk-specific features can enhance the learning experience:

- The app supports *forward progression*, forcing users to only move forward through content. This ensures a predictable learning path and controls interaction duration.
- Users may skip content they are already familiar with in various components.
- A *layered UI* design incorporates a looping screensaver video to attract users when the app is idle.
- A reset option allows users to restart the app and begin from the start.
- If abandoned mid-session, the app starts a countdown and resets after a set period of inactivity, preparing it for the next user.

4 Methodology

4.1 Research design

This study employs a mixed-methods approach to assess user experience and app effectiveness, combining quantitative responses with qualitative, open-ended feedback. The SEESAW app is evaluated with respect to three research questions identified in Section 1: (a) the main usability challenges in designing kiosk-based applications, (b) the effectiveness of the app’s UI in enabling task performance, and (c) how users experience and perceive its usability.

4.2 Sampling process

The SEESAW app was evaluated by 22 participants from multiple domains using purposive sampling. During the evaluation the app was deployed as a web-based application¹, allowing participants to access it from any device. Participants used desktop devices to ensure consistency with kiosk viewports. The participants first received a 10-minute introduction to SEESAW and provided consent to participate². The participants included researchers and university students. They were asked to complete one of two perspectives: *Policy Maker* or *Research Ethics Committee Member*, using their own desktop devices. Both perspectives share similar structure and activities, serving as an experimental constant.

After using the app, participants completed an online questionnaire via Microsoft Forms. The questionnaire first obtained consent, then assessed the app across six sections. These evaluated (1) usefulness as an educational tool, (2) ease of use, (3) learning effectiveness, (4) overall experience, and (5) usability via Nielsen’s heuristics [14], using Likert scale questions. The sixth section included two open-ended questions to capture any usability issues and suggestions for improvement.

4.3 Data analysis

Data from sections 1–5 of the questionnaire is analysed using descriptive statistics, with Likert-scale responses scored from -2 (Strongly disagree) to 2 (Strongly agree). The total and mean scores are calculated to assess each aspect and allow comparisons among individual questions. Finally, qualitative responses are analysed thematically to uncover usability patterns, issues, and suggestions relevant to kiosk-based software.

5 Results

For each of the first four categories (*Usefulness*, *Ease of Use*, *Ease of Learning*, and *User Satisfaction*) and the heuristic evaluation, results are presented as total and average scores per question or heuristic, followed by an overall category average. Each question’s total score is out of 44, and the average ranges from -2 to 2.

5.1 Usefulness

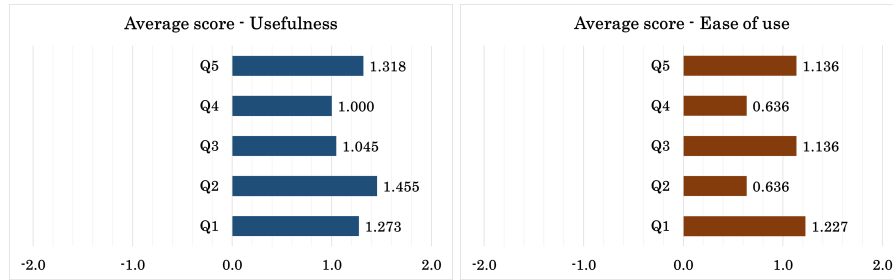
In the Usefulness aspect, participants rated the SEESAW app with an average score of 1.218, indicating a good level of usefulness. As shown in Table 1, users rated the interactive features most highly (1.455), followed by reinforcement

¹ The SEESAW app is accessible as a web app at <https://npaspallis.github.io/seesaw/>

² This research was reviewed and approved by the *Cyprus Bioethics Committee*, Env. No. 2025.01.38

Question	Total score	Average score ([-2, 2])
The SEESAW app helps me better understand research ethics principles.	28	1.273
The interactive features (videos, sorting activities, polls) make learning more engaging.	32	1.455
Using the app has improved my ability to recognize ethical dilemmas in research.	23	1.045
I feel that the app provides relevant information applicable to my role as a student/researcher.	22	1
The SEESAW app would be a useful tool for reinforcing research ethics education.	29	1.318

Table 1: Total and average scores for each question regarding the usefulness of SEESAW.



(a) Average results for each question in the Usefulness section.

(b) Average results for each question in the Ease of use section.

Fig. 5: Results for usefulness and ease of use.

of research ethics (1.318) and understanding principles (1.273). The app also improved users' recognition of ethical dilemmas (1.045) and relevance to their roles (1.0). No participants disagreed with the statements in this section. Figure 5a shows that most users agreed or strongly agreed with the statements.

5.2 Ease of use

Participants found the user interface clear and well-organized (1.227) and understood buttons and gestures easily (1.136). Users were also able to interact without assistance (1.136). However, some were confused during tasks (0.636) and reported that certain instructions were unclear (0.636). These findings are presented in Table 2 and Figure 5b.

5.3 Ease of learning

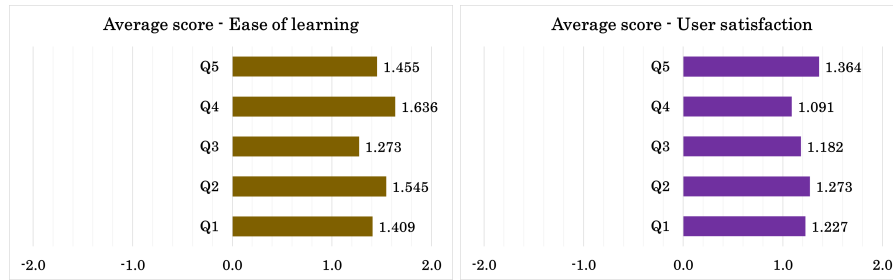
The results in Table 3 and Figure 6a indicate that SEESAW is easy to learn. Participants felt they wouldn't need assistance to reuse it (1.636), that the app has a

Question	Total score	Average score ([-2, 2])
The app's interface is clear and well-organized.	27	1.227
I was able to complete tasks (watch videos, participate in activities, answer polls) without confusion.	14	0.636
The app's buttons, icons, and touch gestures are easy to understand and interact with.	25	1.136
The instructions for each activity were clear and helpful.	14	0.636
I was able to interact with the app without needing external assistance.	25	1.136

Table 2: Total and average scores for each question regarding the ease of use of the SEESAW app.

Question	Total score	Average score ([-2, 2])
I was able to learn how to use the app quickly.	31	1.409
The app provides a logical flow that makes learning easy.	34	1.545
The interactive elements (sorting activities, polls, videos) made it easier to grasp key concepts.	28	1.273
If I were to use the SEESAW app again, I wouldn't need much guidance.	36	1.636
The app's layout and functionality were easy to remember.	32	1.455

Table 3: Total and average scores for each question regarding the learnability of the SEESAW app.



(a) Average results for each question in the Ease of learning section. (b) Average results for each question in terms of User satisfaction.

Fig. 6: Results for learnability and user satisfaction.

logical flow (1.545), and that its layout is memorable (1.455). They also quickly learned to use the app (1.409), with interactive features aiding in understanding (1.273).

5.4 User satisfaction

Question	Total score	Average score ([-2, 2])
I found the SEESAW app enjoyable to use.	27	1.227
The app's design and visuals enhanced my learning experience.	28	1.273
I would use the SEESAW app again to reinforce my understanding of research ethics.	26	1.182
I would recommend the SEESAW app to other students/researchers.	24	1.091
Overall, I am satisfied with my experience using the SEESAW app.	30	1.364

Table 4: Total and average scores for each question regarding user satisfaction.

Participants expressed overall satisfaction with SEESAW (1.227). They enjoyed using the app (1.227), found the design supportive of learning (1.273), and would use it again (1.182). They also reported they would recommend it, though to a slightly lesser extent (1.091). These findings are shown in Table 4 and Figure 6b.

5.5 Heuristic evaluation

Question	Total score	Average score ([-2, 2])
Visibility of system status	27	1.227
Match between system and the real world	32	1.455
User control and freedom	-4	-0.182
Consistency and standards	30	1.364
Error prevention	25	1.136
Recognition rather than recall	24	1.091
Flexibility and efficiency of use	31	1.409
Aesthetic and minimalist design	30	1.364
Help users recognize diagnose and recover from errors	19	0.864
Help and documentation	20	0.909

Table 5: Total and average scores for each heuristic.

The app's overall heuristic score was 1.064 [14]. The highest-rated heuristics were: Match between system and real world (1.455), Flexibility and efficiency

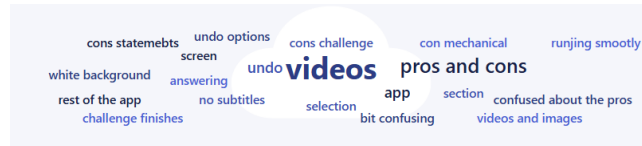
(1.409), and both Consistency and Aesthetic design (1.364). However, participants reported issues with User control and freedom (-0.182). In Heuristic 9 (error recovery), 31% responded as neutral or disagreed. Table ?? contains the full results.

5.6 Qualitative feedback

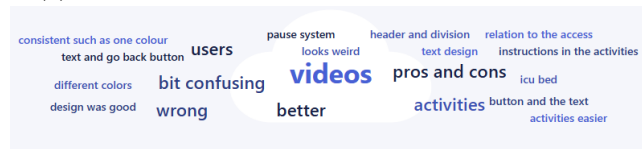
Open-ended responses revealed key usability concerns and suggestions for improvement. The participants' suggestions are taken into consideration to improve the usability of the SEESAW app and enhance its features. Coupled with the results from the previous stage of the evaluation and the challenges encountered during the development process, these can provide a more holistic perspective to address the research questions.

Issues encountered Figure 7a highlights key issues, notably with “videos” and the “pros and cons” activity. Users reported frustration over video playback controls and lack of subtitles. Others found the sorting activity confusing, especially the inability to undo placements. One participant noted the credits screen lasted for too long and they could not exit back to the start screen.

Suggestions Suggestions aligned with the issues identified (Figure 7b). Users proposed consistent colours for draggable items, captions and playback controls for videos, and better responsiveness to enhance UI consistency, perhaps indicating that they encountered some inconsistencies in terms of how the user interface was displayed.



(a) Problems encountered by participants in SEESAW.



(b) Participant suggestions for SEESAW.

Fig. 7: Word clouds for problems faced and suggestions for SEESAW.

6 Analysis & Discussion

The results in Section 5 are analyzed through descriptive statistics and qualitative insights to address the three research questions. The Usefulness of the app averaged 1.218, suggesting that participants found it a helpful educational tool for learning research ethics. High scores were given for the engaging nature of the interactive activities (1.455) and the app’s educational effectiveness (1.318). Ease of use’ was slightly lower at 0.955, indicating that some tasks were confusing. However, most participants agreed that the interface was clear (1.227) and controls were intuitive (1.136). The app’s learnability was high (1.46) suggesting that users quickly understood its functionality (1.409) and needed no guidance to reuse it (1.636). Overall satisfaction was 1.227, with participants finding the app enjoyable and likely to recommend it (1.091). This is reinforced by Qualitative feedback which supported these results. SEESAW was generally seen as helpful and engaging, despite the issues with video and sorting activities.

6.1 Addressing the research questions

RQ1: What are the main usability challenges in designing educational kiosk-based applications? The most prominent challenge identified is the lack of control (H3), affecting two activities in SEESAW. First, the video activity lacks playback controls such as pause, seek, and mute. Second, the pros and cons activity does not allow users to undo placements or move items between boxes. This issue extends across the app, which employs a unidirectional flow design. While intended to guide users through content, this rigid flow limits navigation flexibility and negatively impacts user experience. Participants noted the inability to navigate back and forth, which hindered actions like rewatching videos or revisiting prompts. Although this flow may be necessary to preserve poll integrity, a more flexible approach could improve usability in other areas.

RQ2: How effective is SEESAW’s User Interface (UI) in enabling a good level of task performance? The results indicate that SEESAW’s user interface supports a high level of task performance. Participants found the interface clear and well-organised, contributing to effective task completion. UI elements were easy to learn and use, enabling users to interact efficiently with the app. Optional content skipping allowed users to progress through tasks more quickly, benefiting both novices and those already familiar with the subject. However, improvements are needed in some interactive components, particularly drag-and-drop gestures.

RQ3: How do users experience the SEESAW app, and how do they perceive its usability? Overall, participants reported a very positive experience with SEESAW. Most agreed or strongly agreed with statements about the app’s usability, describing it as pleasant, visually appealing, and engaging. The interactive activities contributed significantly to user satisfaction. However,

feedback on specific components suggests that this high usability may not be consistent across all parts of the app.

7 Conclusions

This study introduced SEESAW, a kiosk-based mobile educational app aimed at students and researchers, focusing on research ethics and the complex decisions researchers face. It uses two scenarios from different perspectives and various interactive pages. The paper outlines UI decisions made during development, internal components, and design considerations for kiosk deployment. SEESAW was evaluated in terms of *Usefulness*, *Satisfaction*, and *Ease of Use* (USE), as well as through Nielsen’s heuristics [15], using both quantitative and qualitative participant feedback. Results were generally positive across all usability dimensions, indicating the app is effective, easy to use, and engaging through its structured flow and interactivity. It encouraged users to complete the experience and reinforced learning outcomes, but some limitations were identified, particularly regarding user control and error recovery—key concerns in kiosk-based design. Issues like limited video controls and the inability to undo navigation or sorting actions caused occasional confusion among participants.

In the future, we plan to expand this study to receive feedback from a larger set and variety of participants. Furthermore, our experiment could be conducted in a more realistic setting, where participants utilize the app under a larger variety of environments and in different contexts. This would increase the potential of uncovering various types of issues when it comes to evaluating the usability of kiosk-based applications and provide further insights.

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